```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
table1 #está no formato tidy, pois, cada variável ocupa uma coluna distinta e cada valor tem sua própia
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <dbl>      <dbl>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```

```r
table2 #não está no formato tidy, pois, a coluna type possuí o nome de variáveis e a coluna count possu
```

```
## # A tibble: 12 x 4
##    country      year type            count
##    <chr>       <dbl> <chr>           <dbl>
##  1 Afghanistan  1999 cases             745
##  2 Afghanistan  1999 population   19987071
##  3 Afghanistan  2000 cases            2666
##  4 Afghanistan  2000 population   20595360
##  5 Brazil       1999 cases           37737
##  6 Brazil       1999 population  172006362
##  7 Brazil       2000 cases           80488
##  8 Brazil       2000 population  174504898
##  9 China        1999 cases          212258
## 10 China        1999 population 1272915272
## 11 China        2000 cases          213766
## 12 China        2000 population 1280428583
```

```r
table3 #não está no formato tidy, pois, a coluna rate possuí duas variáveis distintas, combinadas na me
```

```
## # A tibble: 6 x 3
##   country      year rate
##   <chr>       <dbl> <chr>
## 1 Afghanistan  1999 745/19987071
## 2 Afghanistan  2000 2666/20595360
## 3 Brazil       1999 37737/172006362
## 4 Brazil       2000 80488/174504898
## 5 China        1999 212258/1272915272
## 6 China        2000 213766/1280428583
```

```r
table4a#não está no formato tidy, pois, as colunas(1999,2000) representariam os valores de uma variável
```

```
## # A tibble: 3 x 3
##   country     `1999` `2000`
##   <chr>        <dbl>  <dbl>
## 1 Afghanistan    745   2666
## 2 Brazil       37737  80488
## 3 China       212258 213766
```

```r
table4b#não está no formato tidy, pois, possuí o mesmo caso da table4b.
```

```
## # A tibble: 3 x 3
##   country         `1999`     `2000`
##   <chr>            <dbl>      <dbl>
## 1 Afghanistan   19987071   20595360
## 2 Brazil       172006362  174504898
## 3 China       1272915272 1280428583
```

```r
library(dplyr)
```

```r
taxas <- table1 %>%
  mutate(taxa =(cases / population) * 10000)
taxas
```

```
## # A tibble: 6 x 5
##   country      year  cases population  taxa
##   <chr>       <dbl>  <dbl>      <dbl> <dbl>
## 1 Afghanistan  1999    745   19987071 0.373
## 2 Afghanistan  2000   2666   20595360 1.29
## 3 Brazil       1999  37737  172006362 2.19
## 4 Brazil       2000  80488  174504898 4.61
## 5 China        1999 212258 1272915272 1.67
## 6 China        2000 213766 1280428583 1.67
```

```r
casos_por_ano <- table1 %>%
  group_by(year) %>%
  summarise(total_casos_ano = sum(cases))
casos_por_ano
```

```
## # A tibble: 2 x 2
##    year total_casos_ano
##   <dbl>           <dbl>
## 1  1999          250740
## 2  2000          296920
```

```r
casos_por_pais <- table1 %>%
  group_by(country) %>%
  summarise(total_casos_país = sum(cases))
casos_por_pais
```

```
## # A tibble: 3 x 2
##   country     total_casos_país
##   <chr>                  <dbl>
## 1 Afghanistan             3411
## 2 Brazil                118225
## 3 China                 426024
```

```r
mudanca_casos <- table1 %>%
  filter(year %in% c(1999,2000)) %>%
  group_by(country, year) %>%
  summarise(total_casos = sum(cases))
```

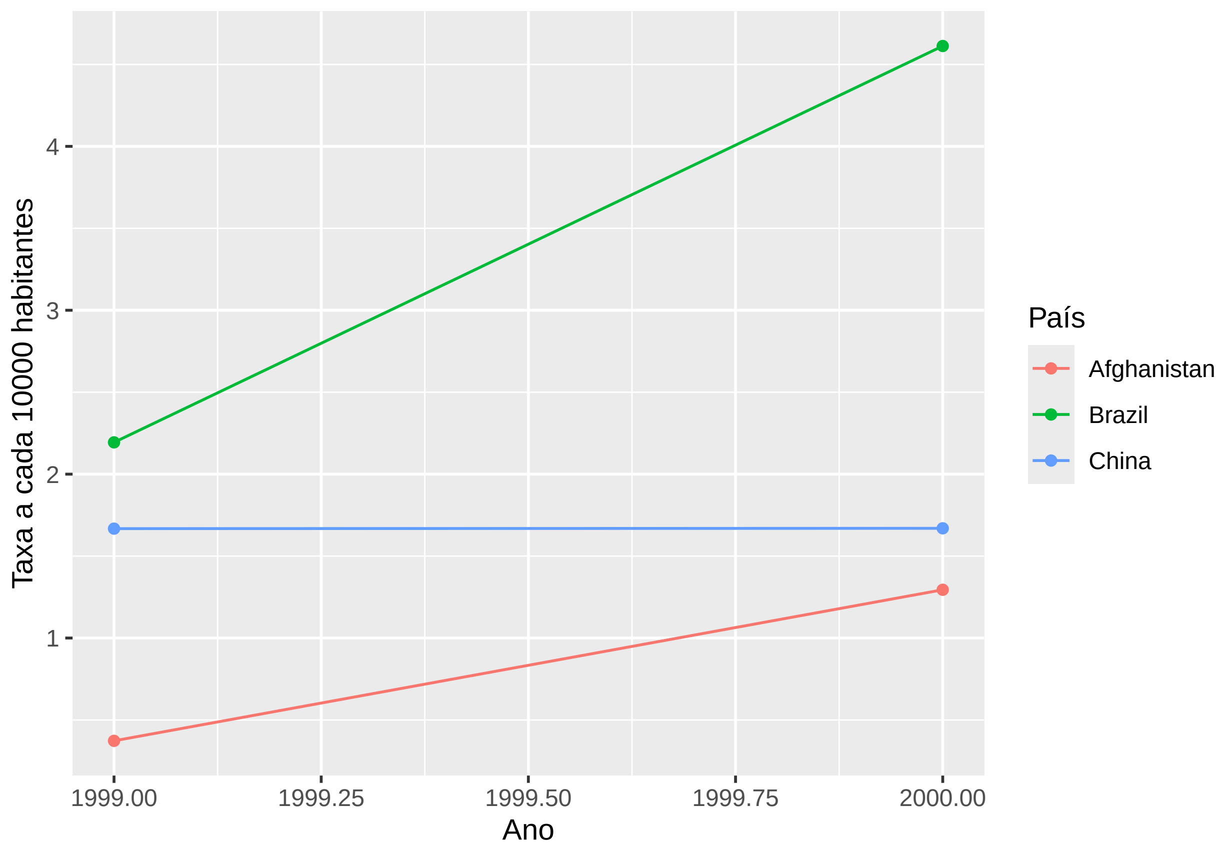
```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by country and year.
## i Output is grouped by country.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(country, year))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
mudanca_casos
```

```
## # A tibble: 6 x 3
## # Groups:   country [3]
##   country      year total_casos
##   <chr>       <dbl>       <dbl>
## 1 Afghanistan  1999         745
## 2 Afghanistan  2000        2666
## 3 Brazil       1999       37737
## 4 Brazil       2000       80488
## 5 China        1999      212258
## 6 China        2000      213766
```

```r
library(ggplot2)
```

```r
#taxas -> Reutilizando da questão
ggplot(taxas, aes(x= year, y=taxa, colour = country, group = country)) +
  geom_line() +
  geom_point() +
  labs( x = "Ano", y = "Taxa a cada 10000 habitantes", color = "País")
```




```r
table2_dados <- table2 %>%
  pivot_wider(names_from = type, values_from = count)
table2_dados
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <dbl>      <dbl>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```

```r
table2_taxas <- table2_dados %>%
  mutate(taxa = (cases/population) * 10000)
table2_taxas
```

```
## # A tibble: 6 x 5
##   country      year  cases population  taxa
##   <chr>       <dbl>  <dbl>      <dbl> <dbl>
## 1 Afghanistan  1999    745   19987071 0.373
## 2 Afghanistan  2000   2666   20595360 1.29
## 3 Brazil       1999  37737  172006362 2.19
## 4 Brazil       2000  80488  174504898 4.61
## 5 China        1999 212258 1272915272 1.67
## 6 China        2000 213766 1280428583 1.67
```

```r
table4a_casos <- table4a %>%
  pivot_longer(cols = c('1999', '2000'), names_to = "year", values_to = "cases")
table4a_casos
```

```
## # A tibble: 6 x 3
##   country     year   cases
##   <chr>       <chr>  <dbl>
## 1 Afghanistan 1999     745
## 2 Afghanistan 2000    2666
## 3 Brazil      1999   37737
## 4 Brazil      2000   80488
## 5 China       1999  212258
## 6 China       2000  213766
```

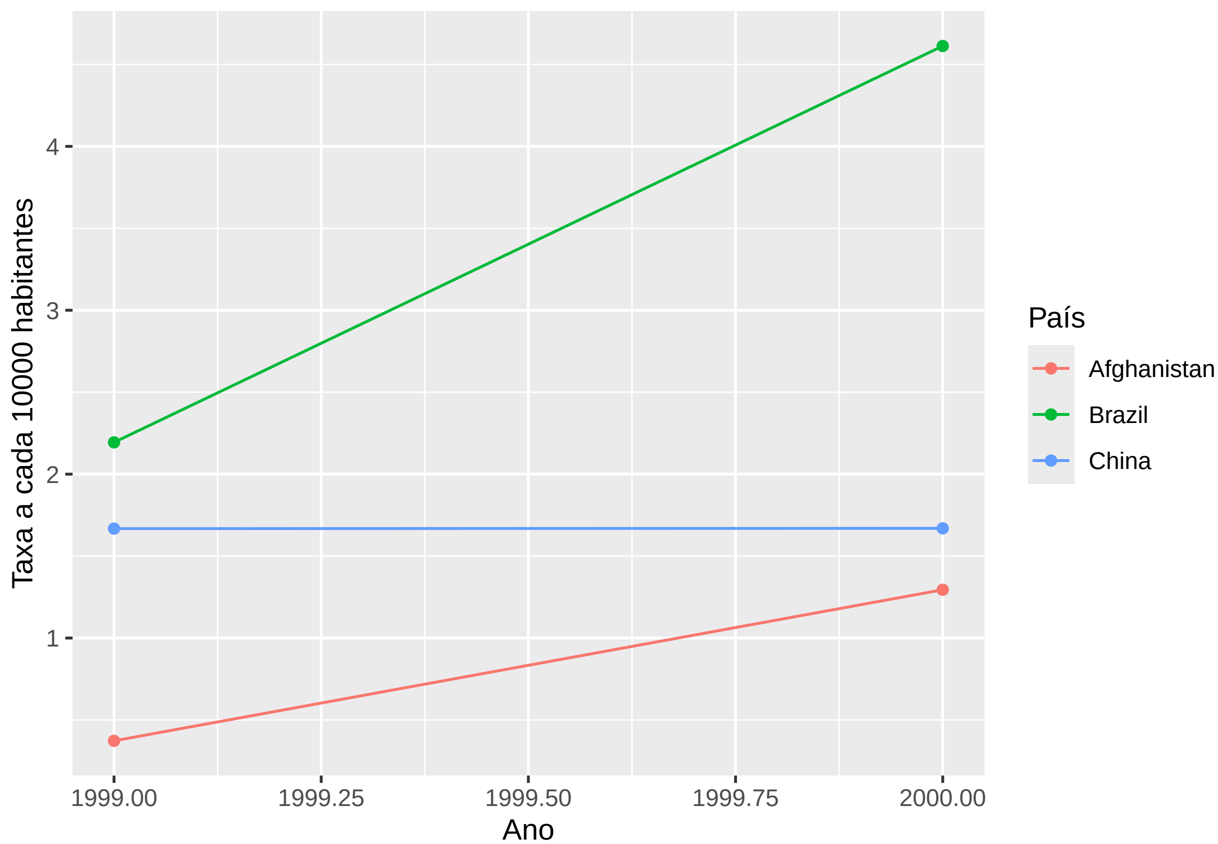
```r
table4b_populacao <- table4b %>%
  pivot_longer(cols = c('1999', '2000'), names_to = "year", values_to = "population")
table4b_populacao
```

```
## # A tibble: 6 x 3
##   country     year  population
##   <chr>       <chr>      <dbl>
## 1 Afghanistan 1999    19987071
## 2 Afghanistan 2000    20595360
## 3 Brazil      1999   172006362
## 4 Brazil      2000   174504898
## 5 China       1999  1272915272
## 6 China       2000  1280428583
```

```r
table4_taxas <- left_join(table4a_casos, table4b_populacao, by = c("country", "year")) %>%
  mutate(taxa = (cases/population)* 10000)
table4_taxas
```

```
## # A tibble: 6 x 5
##   country     year   cases population  taxa
##   <chr>       <chr>  <dbl>      <dbl> <dbl>
## 1 Afghanistan 1999     745   19987071 0.373
## 2 Afghanistan 2000    2666   20595360 1.29
## 3 Brazil      1999   37737  172006362 2.19
## 4 Brazil      2000   80488  174504898 4.61
## 5 China       1999  212258 1272915272 1.67
## 6 China       2000  213766 1280428583 1.67
```

```r
ggplot(table2_taxas, aes(x= year, y=taxa, colour = country, group = country)) +
  geom_line() +
  geom_point() +
  labs( x = "Ano", y = "Taxa a cada 10000 habitantes", color = "País")
```




```r
#usando da questão 8_3
tidy4a <- table4a %>%
  pivot_longer(cols = c('1999', '2000'), names_to = "year", values_to = "cases")
tidy4a
```

```
## # A tibble: 6 x 3
##   country     year   cases
##   <chr>       <chr>  <dbl>
## 1 Afghanistan 1999     745
## 2 Afghanistan 2000    2666
## 3 Brazil      1999   37737
## 4 Brazil      2000   80488
## 5 China       1999  212258
## 6 China       2000  213766
```

```r
#novamente, usando da questão 8_3
tidy4b <- table4b %>%
  pivot_longer(cols = c('1999', '2000'), names_to = "year", values_to = "population")
tidy4b
```

```
## # A tibble: 6 x 3
##   country     year  population
##   <chr>       <chr>      <dbl>
## 1 Afghanistan 1999    19987071
## 2 Afghanistan 2000    20595360
## 3 Brazil      1999   172006362
## 4 Brazil      2000   174504898
## 5 China       1999  1272915272
```

```
## 6 China       2000  1280428583
```

```r
#usando da questão 8_4 agora
tidy4ab <- left_join(tidy4a, tidy4b, by = c("country", "year")) %>%
  mutate(taxa = (cases/population)* 10000)
tidy4ab
```

```
## # A tibble: 6 x 5
##   country     year   cases population  taxa
##   <chr>       <chr>  <dbl>      <dbl> <dbl>
## 1 Afghanistan 1999     745   19987071 0.373
## 2 Afghanistan 2000    2666   20595360 1.29
## 3 Brazil      1999   37737  172006362 2.19
## 4 Brazil      2000   80488  174504898 4.61
## 5 China       1999  212258 1272915272 1.67
## 6 China       2000  213766 1280428583 1.67
```

```r
"O comando left_join serve para unir duas tabelas diferentes em uma só, usando colunas em comum como uma

No exercício, como o número de casos estava na tidy4a e a população na tidy4b, o left_join foi necessári
```

```
## [1] "O comando left_join serve para unir duas tabelas diferentes em uma só, usando colunas em comum c
```

```r
#utilizando da questão 8_1
tidy2 <- table2 %>%
  pivot_wider(names_from = type, values_from = count)
tidy2
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <dbl>      <dbl>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```

```r
tidy3 <- table3 %>%
  separate(rate, into = c("cases", "population"), sep = "/", convert = TRUE)
tidy3
```

```
## # A tibble: 6 x 4
##   country      year  cases population
##   <chr>       <dbl>  <int>      <int>
## 1 Afghanistan  1999    745   19987071
## 2 Afghanistan  2000   2666   20595360
## 3 Brazil       1999  37737  172006362
## 4 Brazil       2000  80488  174504898
## 5 China        1999 212258 1272915272
## 6 China        2000 213766 1280428583
```